Cost-Optimal Sourcing of Industrial Diamonds for Drill Manufacturing

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Abstract - This study formulates a linear programming model for purchasing industrial diamonds used in heavy-, medium-, and light-duty drills. Three suppliers offer different prices and quality mixtures. A Gurobi solution minimizes cost while satisfying all production requirements. The optimal plan purchases 1,647.06 kg from Supplier 1 and 3,470.59 kg from Supplier 3, with no purchase from Supplier 2, for a minimum cost of $14,311.76. The result shows that the lowest quoted price does not necessarily produce the lowest feasible system cost.

Keywords: *linear programming; procurement; supplier selection; Gurobi; cost minimization*

1. Introduction

Manufacturers often buy materials that differ in both price and quality composition. Selecting the cheapest supplier without considering production requirements can increase total cost. This project examines a company that purchases industrial diamonds for three drill categories. The objective is to determine purchase quantities from three suppliers that meet all material requirements at minimum cost. Linear programming is suitable because cost and quality contributions are proportional to purchased quantities [1], [2].

2. Problem Data

The company requires 1,700 kg of heavy-duty-suitable diamonds, 1,200 kg of medium-duty material, and 1,800 kg of light-duty material. Supplier prices and quality fractions are summarized below.

Table 1. Supplier cost and quality composition.

Supplier	Cost ($/kg)	Heavy	Medium	Light
1	3.00	40%	35%	25%
2	2.50	35%	55%	10%
3	2.70	30%	30%	40%

The model assumes constant prices, deterministic quality fractions, divisible purchases, unlimited supplier capacity, and no fixed ordering or transportation costs.

3. Mathematical Formulation

Let x1, x2, and x3 be kilograms purchased from Suppliers 1, 2, and 3. The model is:

$$\text{Minimize } Z = 3.00x1 + 2.50x2 + 2.70x3$$
$$0.40x1 + 0.35x2 + 0.30x3 \geq 1{,}700 \quad \text{(heavy-duty)}$$
$$0.35x1 + 0.55x2 + 0.30x3 \geq 1{,}200 \quad \text{(medium-duty)}$$
$$0.25x1 + 0.10x2 + 0.40x3 \geq 1{,}800 \quad \text{(light-duty)}$$
$$x1, x2, x3 \geq 0$$

4. Gurobi Implementation

The data-driven Python implementation avoids writing a separate constraint for every category:

```
import gurobipy as gp
from gurobipy import GRB
S = [1, 2, 3]; G = ["Heavy", "Medium", "Light"]
cost = {1:3.00, 2:2.50, 3:2.70}
quality = {(1,"Heavy"):0.40, (1,"Medium"):0.35, (1,"Light"):0.25,
           (2,"Heavy"):0.35, (2,"Medium"):0.55, (2,"Light"):0.10,
           (3,"Heavy"):0.30, (3,"Medium"):0.30, (3,"Light"):0.40}
required = {"Heavy":1700, "Medium":1200, "Light":1800}
m = gp.Model("diamond_drill_procurement")
x = m.addVars(S, lb=0, vtype=GRB.CONTINUOUS, name="purchase_kg")
m.setObjective(gp.quicksum(cost[s]*x[s] for s in S), GRB.MINIMIZE)
m.addConstrs((gp.quicksum(quality[s,g]*x[s] for s in S)
              >= required[g] for g in G), name="requirement")
m.optimize()
if m.Status == GRB.OPTIMAL:
    for s in S: print(f"Supplier {s}: {x[s].X:,.2f} kg")
    print(f"Minimum cost: ${m.ObjVal:,.2f}")
```

5. Optimal Results

Table 2. Optimal purchase plan and cost.

Supplier	Purchase (kg)	Cost contribution ($)
Supplier 1	1,647.06	4,941.18
Supplier 2	0.00	0.00
Supplier 3	3,470.59	9,370.59
Total	5,117.65	14,311.76

6. Verification and Interpretation

Table 3. Requirement satisfaction at the optimum.

Category	Available (kg)	Required (kg)	Slack (kg)
Heavy-duty	1,700.00	1,700.00	0.00
Medium-duty	1,617.65	1,200.00	417.65
Light-duty	1,800.00	1,800.00	0.00

Heavy- and light-duty constraints are binding, while the medium-duty requirement has 417.65 kg of slack. Supplier 2 is excluded despite its $2.50/kg price because only 10% of its material is light-duty suitable. Supplier 3 supplies that binding need efficiently, and Supplier 1 provides the complementary heavy-duty mix.

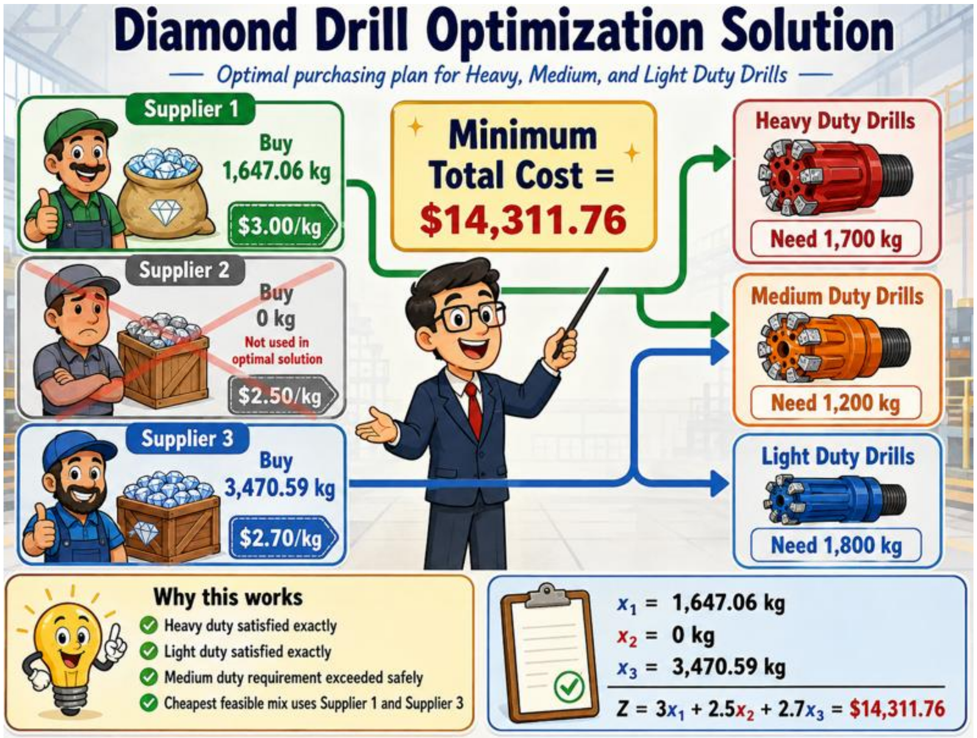

Figure 1. Visual summary of the optimal purchasing plan.

7. Sensitivity and Managerial Implications

At the current optimal basis, an additional kilogram of heavy-duty requirement increases minimum cost by about \$6.176, an additional kilogram of light-duty requirement adds about \$2.118, and the medium-duty marginal value is zero because that constraint is nonbinding. Supplier 2 has a reduced cost of approximately \$0.126/kg; its price would need to fall below about \$2.374/kg before it becomes competitive under the current assumptions. These values identify the specifications of driving cost and provide useful negotiation benchmarks.

8. Limitations and Extensions

The model assumes certain quality fractions, unlimited capacity, continuous quantities, and no logistics, fixed ordering, inventory, or supplier-risk costs. A practical extension could include supplier capacities, minimum orders, binary supplier-selection variables, transportation charges, multiple planning periods, and robust or stochastic constraints for uncertain quality.

9. Conclusion

The linear program identifies a minimum-cost purchase of 1,647.06 kg from Supplier 1 and 3,470.59 kg from Supplier 3, with no purchase from Supplier 2. The resulting cost is \$14,311.76. The key lesson is that supplier price must be evaluated together with material composition and production constraints. Gurobi provides a transparent and reusable way to convert these trade-offs into a defensible procurement decision.

References

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